

**AN INTELLIGENT SYSTEM FOR DETECTION,
TRANSLATION, AND HISTORICAL MAPPING OF
ANCIENT STONE INSCRIPTIONS FROM THE
ANURADHAPURA KINGDOM SRILANKA**

25-26J-144

Project Proposal Report
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August 2025

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ABSTRACT

This component focuses on developing a robust AI-driven system for character segmentation and recognition of ancient Brahmi script inscriptions from the Anuradhapura Kingdom that have suffered physical degradation and erosion over centuries. The research addresses challenges posed by damaged, faded, and stylistically variant Brahmi characters inscribed on stone, which complicate accurate automated transcription.

The methodology leverages an integrated workflow beginning with image preprocessing to enhance inscription visibility and mask damaged regions. Advanced segmentation approaches, including classical image processing and deep learning-based semantic segmentation models, are utilized to isolate individual Brahmi glyphs from complex stone images. Subsequently, convolutional neural networks (CNNs) trained on extensive, annotated Brahmi character datasets perform recognition, even for incomplete or partially missing characters.

A novel feature is the inclusion of uncertainty quantification to flag ambiguous or low-confidence recognitions, providing a human-in-the-loop mechanism for review and correction. The outputs are digital text sequences of Brahmi characters annotated by recognition confidence, laying the foundation for downstream translation and restoration modules.

This work promises to significantly accelerate and improve the reliability of epigraphic transcription efforts by introducing scalable, high-accuracy AI tools specialized for Sri Lanka's ancient inscriptions. It facilitates the preservation, documentation, and subsequent semantic analysis of invaluable cultural heritage artifacts.

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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TABLE OF CONTENT

ABSTRACT	i
LIST OF TABLES	iv
LIST OF FIGURES	iv
LIST OF ABBREVIATIONS	iv
1. INTRODUCTION	1
1.1 Background	1
1.2 Research Gap	2
1.3 Literature Survey	4
1.4 Research Problem	6
2. OBJECTIVES	8
2.1 Main Objective	8
2.2 Specific Objectives	9
3. METHODOLOGY	11
3.1 Overall System Architecture	11
3.2 Character Segmentation and Recognition	13
4. PROJECT REQUIREMENTS	16
4.1 Functional Requirements	16
4.2 Non-Functional Requirements	17
5. BUDGET & BUDGET JUSTIFICATION	20
Justifications:	21
References	22

LIST OF TABLES

Table 1: Comparison between existing systems	04
Table 2: Test Cases	23
Table 3: Budget Details	25

LIST OF FIGURES

Figure 1: Overall System Architecture Diagram	13
Figure 2: Overall Component Diagram	16

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BCE	Before Common Era
CNN	Convolutional Neural Network
KG	Knowledge Graph
ML	Machine Learning
NLP	Natural Language Processing
NER	Named Entity Recognition
OCR	Optical Character Recognition
RDF	Resource Description Framework
GPU	Graphics Processing Unit
API	Application Programming Interface

1. INTRODUCTION

1.1 Background

The Brahmi script is recognized as one of the earliest writing systems in South Asia, dating back to approximately the 3rd century BCE. It laid the foundation for many modern South Asian scripts, including Sinhala, the principal language of Sri Lanka. Inscriptions carved in Brahmi script on stone pillars, slabs, and temple walls constitute a crucial primary source for reconstructing the socio-political, religious, and linguistic history of the Anuradhapura Kingdom, which thrived from the 4th century BCE to the early medieval period.

These inscriptions document royal decrees, land grants, religious donations, and administrative transactions that provide valuable insights into early Sri Lankan civilization. However, centuries of exposure to the elements have caused significant erosion, cracking, and fading of these inscriptions. As a result, many inscriptions are partially illegible, fragmented, or missing entire sections, creating a considerable challenge for epigraphists and historians aiming to transcribe and interpret their content.

The manual deciphering of these inscriptions is a labor-intensive process requiring deep expertise in palaeography, linguistics, and archaeology. It involves meticulous visual inspection of photographs or rubbings and comparison with known glyph variants. This method is slow, prone to human error, and limited by the availability of experts. Moreover, the unique stylistic and regional variations in the Brahmi script and the frequent presence of damaged texts compound the difficulty of accurate transcription.

1.2 Research Gap

Despite encouraging advancements, substantial gaps persist in realizing fully integrated, robust pipelines tailored specifically for the Sri Lankan Anuradhapura Brahmi corpus. Most existing solutions treat OCR, transliteration, and translation as sequential, isolated steps. This limits interoperability and prevents the implementation of contextual feedback mechanisms that could resolve ambiguities caused by inscription damage or linguistic nuances. Consequently, these modular disconnects result in incomplete semantic structuring and hamper the extraction of complex, query able historical knowledge graphs specific to these inscriptions.

Furthermore, although state-of-the-art NLP techniques such as dynamic topic modeling, contextual embeddings, and knowledge graph embeddings have demonstrated impressive capabilities in contemporary domains, their adoption for ancient scripts and minority languages remains limited. Sri Lankan Brahmi and Old Sinhala possess unique linguistic and cultural characteristics that are inadequately captured by general-purpose NLP models, which are predominantly trained on modern and widely spoken languages. The scarcity of domain-specific, well-annotated datasets embodying script variations and inscription-specific semantic concepts hampers effective training and fine-tuning of AI models.

Moreover, there is a distinct lack of comprehensive methodologies that dynamically integrate restoration feedback loops linking AI-driven image and text repair directly with translation and semantic analysis workflows. Such integration is critical to managing fractured and incomplete inscription data in a principled way. Without these feedback loops, the opportunity to iteratively enhance translation fidelity and semantic richness through mutual information exchange among components remains unrealized.

Therefore, there is a pressing research need to develop an end-to-end AI-powered system tailored for the Anuradhapura Brahmi inscriptions. This system should interweave CNN-based script recognition, linguistic and cultural translation expertise, advanced NLP-driven knowledge extraction, and iterative restoration

feedback mechanisms. Such an approach would produce accurate bilingual translations alongside richly structured, semantics-aware historical knowledge graphs that enable sophisticated scholarly analysis and support digital heritage preservation.

Paper 01

Heenkenda, H. M. S. C. R., & Fernando, T. G. I. (2023). *Chronological attribution of Sinhalese inscriptions using deep learning techniques.* Journal of National Science Foundation of Sri Lanka, 51(3).

Discusses deep learning classification on Sinhalese inscriptions including Brahmi script variations in Sri Lanka with data augmentation and transfer learning.

Paper 02

Bandara, P. (2014). *Computational framework for accurate character recognition of early Brahmi scripts of Sri Lanka.*

Proposes an image processing approach combined with recognition methods tailored for Sri Lankan Brahmi characters.

Paper 03

Ruwanmini, S., Dias, K., Niluckshini, C., & Nandasara, T. (2016). *Sinhala inscription character recognition model using neural networks.*

Reports a system combining OCR and GIS tools specializing in inscription character recognition integrating classical and AI-based approaches.

Paper 04

Chawla, S., Mahajan, A., & Others (2024). *Deep learning models for ancient script recognition: Tamil and Modi to Brahmi perspective.*

Highlights successes and challenges in applying CNN-based AI models to ancient South Asian scripts similar to Brahmi.

Paper 05

Gautam, A., & et al. (2023). *Advanced AI methods for Brahmi inscription image enhancement and recognition.* Intelligent Systems and Applications in Engineering.

Emphasizes the importance of restoration feedback and domain-specific AI models for erosion-damaged Brahmi inscription digitization.

Table 1: Comparison between existing systems

Feature / Capability	Paper 1	Paper 2	Paper 3	Paper 4	Paper 5	Proposed System
OCR for ancient Brahmi and preprocessing	✓	✓	✓	✓	✓	✓
Use of Deep Learning and CNN models	✓	✗	✓	✓	✓	✓
Restoration feedback loop integration	✗	✗	✗	✗	✓	✓
Uncertainty quantification in recognition	✗	✗	✗	✗	✗	✓

1.3 Literature Survey

Recent decades have witnessed increasing interdisciplinary efforts to apply digital technologies to ancient epigraphy, aiming to automate or assist the transcription process. The convergence of computer vision, machine learning, and natural language processing has enabled the emergence of digital epigraphy as a vibrant research area.

Character Segmentation

Segmentation the step of isolating individual characters or glyphs from inscription images is crucial but challenging. Classical image processing approaches employ edge detection, morphological filtering, and contour analysis to detect character boundaries (Karunaratne & Wijesuriya, 2012). However, these methods often struggle with severely eroded or connected characters.

More recent studies have explored deep learning-based semantic segmentation models like U-Net and Mask R-CNN, which have demonstrated promising results in segmenting degraded text in historical manuscripts (Rathnayake et al., 2024). However, their application to Brahmi inscriptions remains limited due to the paucity of annotated datasets and the unique characteristics of stone inscriptions compared to paper manuscripts.

Optical Character Recognition (OCR)

Traditional OCR systems designed for printed and handwritten modern scripts are ill-equipped to handle ancient Brahmi inscriptions due to stylistic variance and damage. Neural networks, especially convolutional neural networks (CNNs), have revolutionized OCR for complex scripts (Goodfellow et al., 2016), including historical text recognition tasks (Rajasekaran et al., 2023).

For Brahmi, there have been emerging efforts to train CNNs on isolated characters extracted from inscription images, achieving moderate accuracy (Rathnayake et al., 2024). Nevertheless, real-world recognition is hampered by noisy inputs, fragmented glyphs, and non-standard layouts. Combining CNNs with recurrent layers or attention mechanisms has shown potential to model sequential dependencies for continuous text but requires further research tailored to Brahmi.

Uncertainty Quantification

An important but underexplored facet in ancient script recognition is the treatment of uncertainty due to damaged or ambiguous glyphs. Bayesian neural networks and confidence scoring mechanisms have been proposed to indicate recognition reliability, enabling human-

in-the-loop verification (Lakshminarayanan et al., 2017). Incorporating uncertainty is particularly valuable for ancient inscriptions where complete automation is unrealistic.

Integrated Epigraphic Digital Systems

Notable projects such as the “Brahmi to Sinhala Translator” (Hettiarachchi et al., 2020) and the Digital Epigraphy Project (Perera et al., 2019) have combined OCR with linguistic translation and database building for South Asian inscriptions. However, many existing solutions treat segmentation, recognition, and translation as separate modules with limited feedback, reducing the system’s ability to handle fragmented or incomplete texts robustly.

1.4 Research Problem

Despite growing interest and several pioneering efforts in applying AI and machine learning to digital epigraphy, the specific challenges posed by the ancient Brahmi inscriptions of the Anuradhapura Kingdom remain insufficiently addressed. While OCR and image processing technologies have advanced considerably for contemporary and well-preserved historical scripts, Brahmi inscription recognition presents unique obstacles that existing solutions have yet to fully overcome.

Insufficient Annotated Datasets

Machine learning models require large, diverse, and well-annotated datasets for effective training and generalization. Currently, there is a lack of comprehensive, publicly available Brahmi character datasets that capture real-world variation in glyph forms and damage states. Existing datasets tend to be small, experimentally collected, or synthesized, limiting the representativeness of models trained on them.

Limited Specialized Segmentation Methods

Character segmentation for printed or handwritten modern text is a mature field; however, the segmentation of Brahmi glyphs on weathered stone surfaces remains underdeveloped. Classical image processing techniques struggle with fragmented or connected characters, while deep learning segmentation methods have only recently begun to be applied but lack sufficient domain adaptation or annotated training examples tailor-made for Brahmi inscriptions.

Lack of Uncertainty-Aware Recognition

The intermittent and fragmentary nature of Brahmi inscriptions necessitates recognition systems capable of expressing uncertainty. Most OCR engines produce deterministic outputs without confidence quantification, forcing manual reviews blindly. There is a pressing need for models that can identify and flag uncertain or ambiguous glyph recognitions, enabling targeted human-in-the-loop interventions to ensure transcription accuracy.

Integration and Feedback Deficiency

Many current efforts approach segmentation, recognition, and translation as isolated tasks, with limited interoperability or iterative feedback mechanisms. Closed-loop systems where restoration outputs influence recognition, or recognized texts direct image enhancement, remain underexplored in Brahmi context. This limitation hinders progressive improvement of transcription quality critical for damaged inscriptions.

Linguistic and Cultural Context Utilization

Brahmi script recognition benefits from embedding linguistic and cultural knowledge, such as expected glyph sequences, historical vocabulary, and paleographic variants. Current AI models largely neglect this contextual integration, potentially missing cues essential to resolving ambiguous or partially damaged characters.

2. OBJECTIVES

2.1 Main Objective

To develop a robust, AI-driven character segmentation and recognition system specifically designed for the ancient Brahmi stone inscriptions of the Anuradhapura Kingdom, which represent some of the earliest and most significant written records in Sri Lanka, dating back to at least the 3rd century BCE. These inscriptions are invaluable for understanding the political, cultural, religious, and linguistic history of early Sri Lankan civilization, yet centuries of natural erosion, fragmentation, and wear have posed considerable challenges for their manual transcription and study.

This system aims to accurately isolate and recognize degraded, fragmented, and stylistically variable Brahmi glyphs from high-resolution images of these inscriptions, utilizing advanced image processing, deep learning-based segmentation, and convolutional neural network recognition architectures. By producing highly detailed, annotated digital transcriptions with integrated uncertainty quantification, the system will help overcome the limitations of

traditional epigraphic analysis, enabling more efficient and reliable preservation, restoration, and translation workflows.

Ultimately, this project will significantly advance the digitization and scholarly analysis of Sri Lanka's epigraphic heritage, facilitating downstream historical and linguistic research, creating accessible digital archives for cultural education, and contributing to the global understanding of early written communication in South Asia.

2.2 Specific Objectives

Develop Advanced Image Preprocessing Techniques:

Implement and optimize image enhancement methods such as noise reduction, contrast normalization, and damage masking to prepare high-quality inputs from complex and deteriorated inscription images for effective segmentation.

Design and Train Sophisticated Character Segmentation Models:

Create and evaluate both classical image processing pipelines and deep learning-based semantic segmentation networks (e.g., U-Net, Mask R-CNN) tailored to extract accurate character boundaries from noisy, eroded Brahmi inscriptions.

Construct High-Accuracy Character Recognition Models:

Build convolutional neural network (CNN) architectures trained on extensive, annotated Brahmi character datasets, including synthetically augmented damaged variants, to accurately classify individual glyphs under varying levels of degradation.

Incorporate Uncertainty Quantification in Recognition:

Integrate confidence scoring mechanisms and probabilistic models to assess and flag low-confidence or ambiguous character recognitions, allowing targeted human expert review and improving overall transcription quality.

Integrate Segmentation and Recognition into Automated Pipeline:

Develop an end-to-end workflow combining preprocessing, segmentation, and recognition modules that can process batches of inscription images efficiently and output structured digital transcriptions with spatial and confidence annotations.

Validate System Performance Through Rigorous Evaluation:

Employ quantitative metrics such as precision, recall, F1-score for segmentation and recognition, alongside expert epigraphist evaluation, to assess system accuracy, robustness, and practical usability on real inscription datasets.

Prepare Data and Outputs Compatible with Downstream Linguistic and Restoration Tasks:

Ensure that transcription outputs are compatible with modules designed for translation to modern languages and AI-based restoration tools to facilitate complete automated epigraphic analysis.

3. METHODOLOGY

3.1 Overall System Architecture

The workflow for character segmentation and recognition begins with acquiring high-resolution images of Brahmi inscriptions from sites, museums, or archives, using DSLR cameras or mobile phones. These images undergo preprocessing steps including noise removal, contrast adjustment, and damage masking to enhance the visibility of erosion-affected glyphs and mitigate background artifacts.

Following enhancement, character segmentation isolates individual Brahmi symbols from the full inscription image. Segmentation is achieved through a combination of classical image processing methods and deep learning approaches. Projection profile analysis preliminarily segments the text into lines and candidate characters. Morphological filtering and contour detection are applied to refine boundaries. For fragmented or touching glyphs, semantic segmentation models such as U-Net are trained on annotated datasets to accurately separate characters, including those with significant damage or erosion.

Once characters are segmented, a convolutional neural network (CNN)-based classifier is employed to recognize each glyph. The model is trained on an extensive, annotated dataset of Brahmi characters augmented with synthetic noise, occlusion, and deformation to simulate real-world conditions found in ancient inscriptions. Recognition outputs include confidence scores, enabling subsequent uncertainty-aware review.

The system assembles the recognized characters into spatially annotated digital transcriptions. Low-confidence predictions are flagged and forwarded to an expert-in-the-loop interface for manual verification and correction, ensuring accuracy and reliability.

Finally, the transcription data integrates with downstream components for translation and semantic analysis, forming part of an end-to-end pipeline for digital epigraphy and cultural heritage preservation.

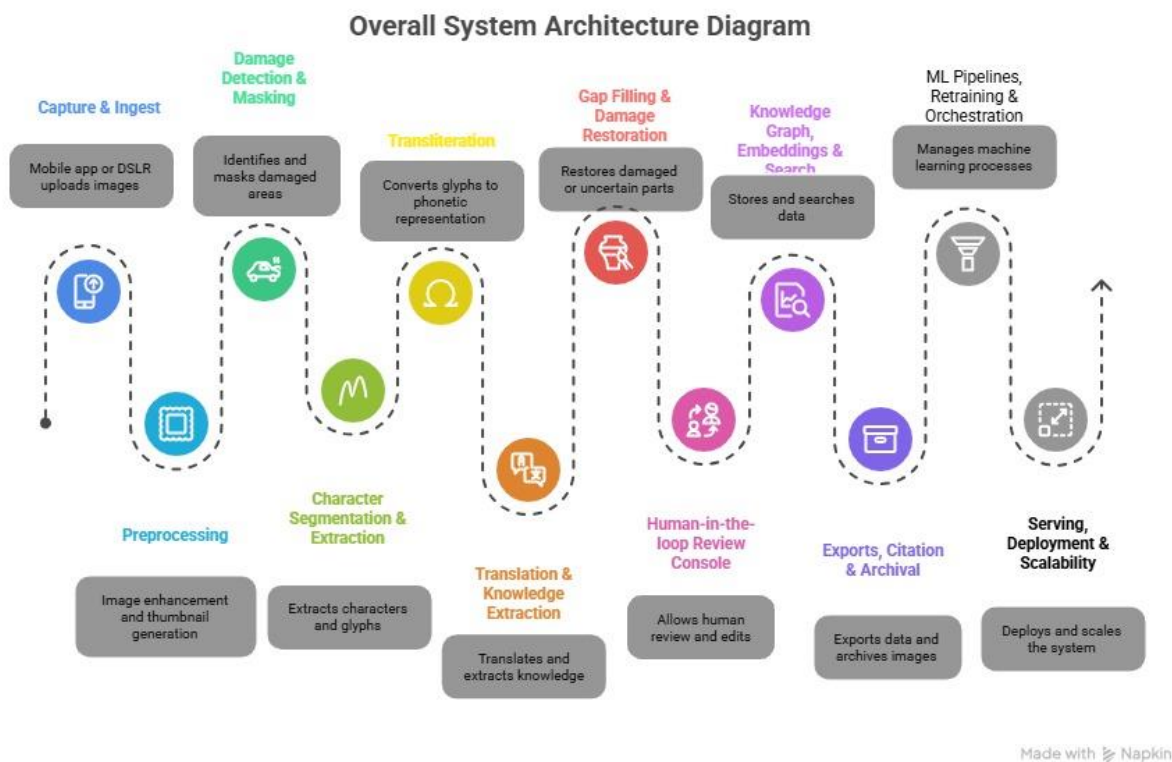


Figure 1: Overall component Diagram

3.2 Character Segmentation and Recognition

This component begins with **preprocessed inscription images** supplied by upstream module.

Character Segmentation:

Projection Profile Analysis: Horizontal and vertical projection profiles segment inscription images into text lines and candidate character bounding boxes.

Classical Image Processing: Edge detection, contour extraction, and morphological filtering separate individual characters and remove noise artifacts.

Deep Learning Semantic Segmentation: U-Net based models trained on labeled Brahmi data further segment touching and degraded characters with pixel-level accuracy, overcoming challenges posed by erosion and overlapping glyph strokes.

Character Recognition:

CNN-Based Classification: Each segmented character image is classified using convolutional neural networks trained on a richly annotated Brahmi glyph dataset augmented with synthetic noise, occlusion, and distortion.

Confidence Scoring: Recognition outputs include softmax probabilities and Bayesian uncertainty measures to flag low-confidence predictions.

Output:

Generates annotated digital transcriptions containing recognized Brahmi characters, spatial location data, and confidence scores.

Human-in-the-Loop Verification:

Confidence-threshold characters are directed to expert reviewers for manual correction, assisting iterative system refinement.

This workflow from preprocessed images to recognized glyphs forms a critical module integrated seamlessly into the broader inscription digitization and semantic analysis pipeline.

Character Segmentation and Recognition

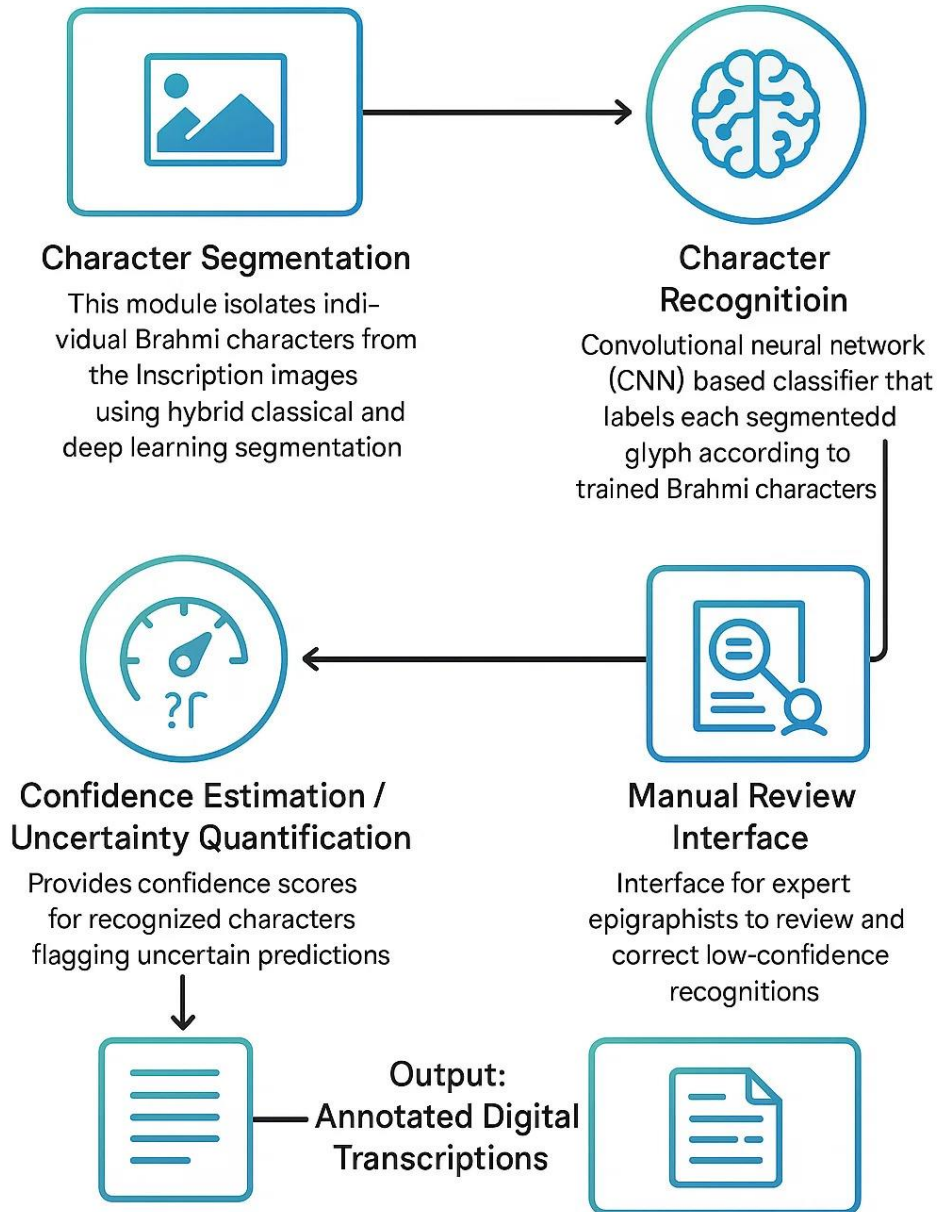


Figure 2: Component diagram Character Identification

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

Image Input Management:

The system shall accept high-resolution digital images of Brahmi stone inscriptions in standard image formats (e.g., JPEG, PNG, TIFF).

It should support batch processing of multiple images.

Image Preprocessing:

The system shall perform noise reduction, contrast adjustment, and damage masking to enhance inscription visibility and clarity prior to segmentation.

Character Segmentation:

The system shall accurately identify and isolate individual Brahmi characters or glyphs from complex stone inscription images, including handling degraded, eroded, or partially missing glyphs.

Segmentation output must provide clear spatial boundaries or masks for each identified character.

Character Recognition:

The system shall classify segmented characters into correct Brahmi glyph categories using trained CNN-based OCR models.

It shall output recognized characters as digital text with associated positional metadata.

Uncertainty Quantification:

The system shall assign confidence scores to each recognized character.

Low-confidence or ambiguous recognitions shall be flagged and made available for manual review.

Output Generation:

The system shall generate annotated digital transcription files containing recognized Brahmi text sequences, spatial information, and confidence annotations.

Output formats shall be compatible with downstream translation, restoration, and knowledge extraction modules.

User Interface for Manual Review:

The system shall provide an interface for epigraphists and linguists to inspect, verify, and correct flagged low-confidence characters.

4.2 Non-Functional Requirements**Accuracy:**

The system shall achieve at least 85% accuracy in character recognition on standard test datasets representing various inscription conditions.

Performance:

The system shall process a batch of 100 inscription images within 8 working hours on standard GPU-enabled hardware.

Scalability and Extensibility:

The architecture shall support incremental retraining and addition of new character classes or annotation data with minimal reengineering.

It shall be modular to allow integration with other inscription analysis components.

Usability:

The user review interface must be intuitive and accessible without deep technical knowledge, supporting easy corrections and annotations.

Reliability and Robustness:

The system shall handle variable image qualities, lighting conditions, and damage patterns gracefully without crashing or significant performance degradation.

Security:

Data confidentiality should be maintained, complying with institutional data protection policies and ethical guidelines concerning heritage material.

Table 2: Test Cases

Test Case ID	Description	Input	Expected Output
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TC001	Preprocess preprocessed Brahmi image input	Preprocessed inscription image	Segmentation-ready image with noise and artefacts minimized
TC002	Character segmentation accuracy	Preprocessed inscription image	Correctly isolated individual Brahmi character images with bounding boxes
TC003	Segmentation of touching or broken characters	Inscription image containing connected/fragme nted glyphs	Properly segmented and separated individual glyph images
TC004	Character recognition on segmented glyphs	Segmented character images	Correct Brahmi character label with confidence scores
TC005	Low-confidence character recognition handling	Segmented glyph images with ambiguous features	Characters flagged for manual expert review
TC006	Output generation	Recognized character labels & locations	Digital transcription with spatial annotation and confidence metadata
TC007	Integration with manual correction interface	Flagged low confidence character images	Post-correction transcription updates
TC008	Robustness under partial inscription data	Images with missing or occluded characters	Reasonable recognition with uncertainty

			indicated and flagged properly
TC009	Batch processing performance	Batch of preprocessed images	Complete segmentation and recognition within acceptable latency per image

5. BUDGET & BUDGET JUSTIFICATION

Table 3: Budget Details

Requirement	Cost (Rs.)
Fieldwork/Travel for Data Collection	Rs. 25,000.00
Purchasing virtual storage	Rs. 4,000.00

Software Licenses (OCR/Image Tools)	Rs. 6,000.00
Miscellaneous (app store fees etc.)	Rs. 5,000.00
Hosting	Rs. 5000.00 per month
Total Cost	Rs. 45,000.00 + monthly hosting fee

Justifications:

- Fieldwork/Travel for Data Capturing (Rs. 25,000.00)

This is for visiting sites and capturing real images of inscriptions. It covers transportation, travel, and modest expenses during data capturing.

- Purchasing Virtual Storage (Rs. 4,000.00)

Since images take up a lot of space, more space is needed to safely store all the data and make it readily available to members of the team.

- Software Licenses (OCR/Image Tools) (Rs. 6,000.00)

Special software will be required for image and text processing. The license cost will allow us to make use of advanced features for accurate results.

- Miscellaneous (Rs. 5,000.00)

This is kept for small or unexpected costs such as app store fees or other minor necessities that might crop up.

- Hosting (Rs. 5,000.00/month)

The system should be hosted online so that it is readily accessible. The monthly subscription will maintain the system operational and always in service.

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